Protocol Interface SAP Live Server version 2.00

Communication

Communication between client and SAP Live server is handled by a standard TCP socket connection. The IP and PORT will be defined by the connection parameter received from Manage2Sail portal. IP and port will not change from one race to another. So it's possible to open more TCP connection in parallel

Basic message frame

```
<STX>Message body<ETX>
```

STX	1 byte	02 Hex
ETX	1 byte	03 Hex

All protocols framed between <STX> und <ETX>.
Requests always have a "?" after the command.

Parameter Legend

- Data between "(" and ")" is repeatable (0-n)
- Data between "[" and "]" is optional
- Data between "{" and "}" is set of possible values
- MessageNr -> unique message number for a single race. Values from 1 to n
- RaceId -> the unique id of the race
- DataCount -> number of following Data
- RaceDescription -> Description of the race
- SailNumber -> the unique boat id
- MarkIndex -> the zero based index of the measurement point (0 is the start)
- MarkType -> the type of the mark (0 – manual, 1 – line crossing, 2 – buoy rounding)
- NOC -> National Olympic Committee Code
- Name -> the name of the Sailor/Boat/Team
- StartTime -> the local start time HH:MM:SS
- MarkTime -> the local day time of the mark rounding HH:MM:SS
- Rank -> the current rank of the boat in the race
- TimeSinceStart -> the time since start HH:MM:SS
- Conditions -> weather conditions

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- Temperature -> temperature in degree Celsius
- Humidity -> humidity in %
- WindDirection -> wind direction in degree
- WindSpeed -> wind speed in knots
- Version -> The version number of the SailMaster

Standard error message for non valid requests: “Invalid command”.
If the command is correct but the parameters are invalid: “Invalid parameter”.
If a request is answered the Command in the answer is followed by a “!”. If this message is sent unrequested, there is no “!” used.
Example:
Request: <STX>RAC?<ETX>
Answer: <STX>RAC!|DataCount…<ETX>

Sent because changes happened or cyclic:
Answer: <STX>RAC|DataCount…<ETX>

1. Command messages

Open Race

- *Sets the current RaceId identifier for this connection. All messages will be then filtered for this race id*
- *It should be the first message in the hand shaking phase*
- *If this value will not be set, no data will be send automatically*
- *Answer to this message will contain last message number, which has been received from tracking system*

```
Client -> LiveTrakingServer
<STX>OPN|RaceId<ETX>

LiveTrackingServer -> Client
<STX>OPN!|{OK:FAILED}[|Last Message Number]<ETX>

Remark: Last Message Number will be sent only if result is “OK”
```

Listen Race

- *By default, sending of live messages is not activated and can be started via this function*
- *Message number parameter indicates the message, from which the live stream should begin*
- *If the Message number parameter is missing, all messages will be send to the client*
- *Race Messages will be send asynchronously after the answer to this command has been send*

```
Client -> LiveTrakingServer
<STX>LSN|{ON:OFF}[|Message number]<ETX>

LiveTrackingServer -> Client
<STX>LSN!|{OK:FAILED}<ETX>
```

Course Configuration

(Automatically sent if changes happen)
(Sorted by MarkIndex)

```
<STX>MessageNr|CCG|RaceId|DataCount(|MarkIndex;MarkDescription;MarkType;
   SailNumber1;SailNumber2)<ETX>
```

Startlist

(Automatically sent if changes happen)

```
<STX>MessageNr|STL|RaceId|DataCount(|SailNumber;NOC;Name)<ETX>
```

Clock at Mark/Finish

(Automatically sent if first boat crosses mark)

```
<STX>MessageNr|CAM|RaceId|DataCount(|MarkIndex;MarkTime;SailNumber)<ETX>
```

Timing Data

(Automatically sent if changes happen)

```
<STX>MessageNr|TMD|RaceId|SailNumber|DataCount(|MarkIndex;Rank;TimeSinceStart)<ETX>
```

Timing Data

```
<STX>MessageNr|RPD|RaceID|RaceStatus|DataTime|StartTime/EstimatedStartTime|RaceTime|
   NextMarkLeader|DistanceToNextMarkLeader|DataCount(|SailNumber;
   BoatType;AgeOfData;Latitude;Longitude;SOG;VMG;ALS;COG;NextMark;Rank;
   DistToLeader;DistToNextMark;BoatIRM)<ETX>
```

Special Parameter

- RaceStatus -> The state of the race (0 – Ready, 1 – Armed, 2 – Running, 3 – Finished) and the RacingStatus (1 - Abandoned,2 - Completed, 3 - Finished, 4 - GeneralRecall, 5 - NoRacingToday, 0 - None, 6 - Postponed, 7 -Racing, 8 - RacingIR, 9 - RunAfter, 10 - StartSequence) e.g. "1,4" , "2,7", "3,3")
- DataTime -> Local time of this snapshot, YYYY-MM-DDTHH:MM:SS+timezone 2011-12-31T12:34:56+01:00
- StartTime/EstimatedStartTime -> Local time, HH:MM:SS
- RaceTime -> Time since start, HH:MM:SS
- NextMarkLeader -> Index of next MeasurementPoint (0 = Start, 1 = First Mark…) for the leading boat.
- DistanceToNextMarkLeader -> the distance to the next mark for the leading Boat in m
- DataCount -> Amount of following Position/Race Information for each competitor and each used official boat/buoy
- BoatType -> (0 – Unidentified, 1 – Competitor, 2 – Committee, 3 – Jury, 4 – TimingScoring, 5 – Buoy)
- AgeOfData -> The age of the data in seconds
- Latitude -> in degree
- Longitude -> in degree
- SOG -> Speed over ground in kn
- ALS -> Average speed over ground per leg in kn
- VMG -> Velocity made good in kn

- COG -> Course over ground in degree
- DistToLeader -> Distance to leader in m
- DistToNextMark -> Distance to next mark in m
- BoatIRM -> The IRM of the boat if given (e.g. DNF, OCS, DNS, BFD…)